

Simple\_LR0.ipynb - Colab

colab.research.google.com/drive/1QOKCDVjKHjfaDgxGyprUzekEM\_AIPUNP#scrollTo=ab8NIZbtqEO7

Simple\_LR0.ipynb

File Edit View Insert Runtime Tools Help All changes saved

+ Code + Text

Comment Share Gemini

RAM Disk

Import the libraries

[46]

import numpy as np  
import matplotlib.pyplot as plt  
import pandas as pd  
from sklearn.linear\_model import LinearRegression

[47]

# Sample data (x=years of experience , y= salary(1000\$))  
x = [1, 2, 3, 4, 5]  
y = [2, 4, 5, 4, 5]

[48]

# Reshape x to be a 2D array as required by scikit-learn  
x = np.array(x).reshape(-1, 1) # Reshape x to a column vector  
y = np.array(y)

[49]

# Create and fit the linear regression model  
model = LinearRegression()  
model.fit(x, y)

LinearRegression  
LinearRegression()

[50]

# Predict y values  
y\_pred = model.predict(x)

completed at 10:30 PM



+ Code + Text



✓  
0s



```
# Coefficients
print("Slope (Coefficient):", model.coef_)
print("Intercept:", model.intercept_)
```



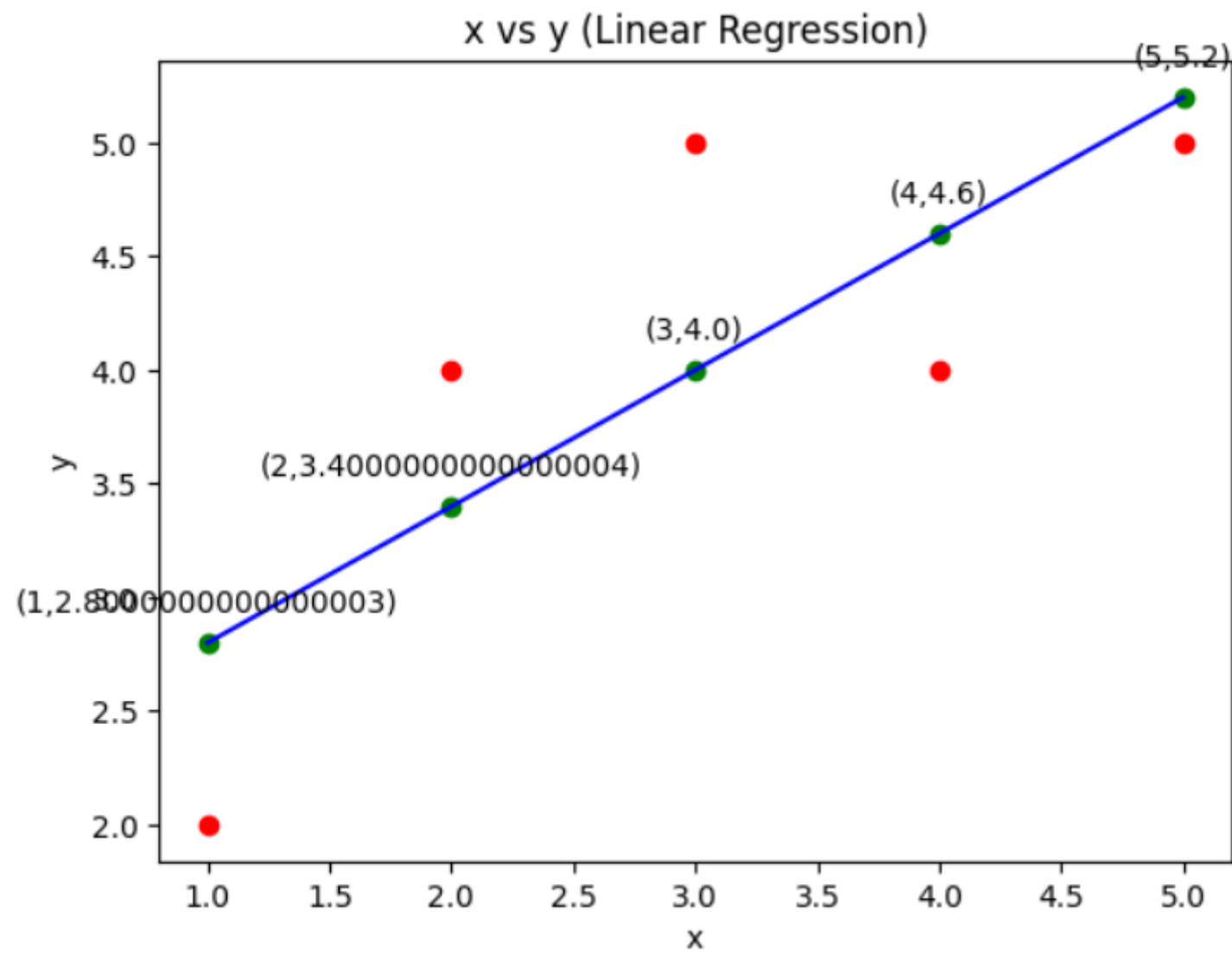
```
Slope (Coefficient): [0.6]
Intercept: 2.2
```



✓  
0s



```
# Plotting the results
plt.scatter(x, y, color='red',label='Actual data') # Original data points
plt.plot(x, y_pred, color='blue',label='Fitted line') # Regression line
plt.scatter(x, y_pred, color='green',marker='o',label='predicted dat')
#Annotate predicted points with coordinates
for i in range(len(x)):
    plt.annotate(f'({x[i][0]},{y_pred[i]})', (x[i][0], y_pred[i]), textcoords="offset points", xytext=(0,10), ha='center')
plt.title('x vs y (Linear Regression)')
plt.xlabel('x')
plt.ylabel('y')
plt.show()
```



### Mean Squared Error (MSE):

1. Compute the difference between each pair of actual and predicted values.
2. Square each difference.
3. Find the average of these squared differences.

✓  
0s

```
# Calculate the Mean Squared Error  
mse = np.mean((y - y_pred) ** 2)  
print(f"Mean Squared Error: {mse}")
```

➞ Mean Squared Error: 0.47999999999999987